

Advanced (Carolina Reaper)

- Q1.** The quadratic function $f(x) = x^2 + 4x + 4$ can be written in vertex form as _____, but it is being discounted by _____.
- (A) $f(x) = (x + 2)^2$ _____ 2
- (B) $f(x) = (x - 2)^2$
- (C) $f(x) = x^2 + 4x + 4$
- (D) $f(x) = (x + 1)^2$
- (E) $f(x) = (x - 1)^2$
- Q2.** A water tank can hold 120 gallons of water. Due to a leak, the tank is losing water at a rate of 2 gallons per minute. The amount of water W in the tank after t minutes can be modeled by the quadratic function $W(t) = -2t^2 + 20t + 100$. Write this function in intercept form.
- (A) $W(t) = -(t - 10)(t - 5)$
- (B) $W(t) = -(t - 10)(t + 5)$
- (C) $W(t) = (t - 10)(t - 5)$
- (D) $W(t) = (t + 10)(t + 5)$
- (E) $W(t) = -(t + 10)(t - 5)$
- Q3.** The vertex form of a quadratic function is $f(x) = (x - 3)^2 + 2$. What is the y -intercept of this function?
- (A) $y = -7$
- (B) $y = -5$
- (C) $y = 2$
- (D) $y = 5$
- (E) $y = 11$
- Q4.** A quadratic function has an axis of symmetry at $x = 2$ and passes through the point $(4, 0)$. Write this function in vertex form.
- (A) $f(x) = (x - 2)^2$
- (B) $f(x) = -(x - 2)^2$
- (C) $f(x) = (x - 2)^2 + 4$
- (D) $f(x) = -(x - 2)^2 + 4$
- (E) $f(x) = (x + 2)^2$
- Q5.** A store is having a sale on a certain item. The original price is _____.
- 20
- Q6.** A quadratic function has x -intercepts at $x = -3$ and $x = 5$. Write this function in intercept form.
- (A) $f(x) = (x + 3)(x + 5)$
- (B) $f(x) = (x + 3)(x - 5)$
- (C) $f(x) = -(x + 3)(x + 5)$
- (D) $f(x) = (x - 3)(x - 5)$
- (E) $f(x) = -(x - 3)(x - 5)$
- Q7.** A ball is thrown from the top of a building. The height h of the ball after t seconds is given by the quadratic function $h(t) = -4t^2 + 20t + 10$. What is the maximum height of the ball?
- (A) 20 feet
- (B) 25 feet
- (C) 30 feet
- (D) 35 feet
- (E) 40 feet
- Q8.** The quadratic function $f(x) = x^2 - 6x + 8$ can be written in vertex form as _____.
- (A) $f(x) = (x - 3)^2 - 1$
- (B) $f(x) = (x - 3)^2 + 1$
- (C) $f(x) = (x + 3)^2 - 1$
- (D) $f(x) = (x - 2)^2 - 4$
- (E) $f(x) = (x + 2)^2 - 4$

Open Ended Questions (FRQ)

- Q9.** A company produces a product with a cost function $C(x) = 2x^2 + 10x + 50$, where x is the number of units produced. Write this function in vertex form and find the minimum cost.
- Q10.** A water park has a wave pool with a depth function $d(x) = -x^2 + 10x + 5$, where x is the distance from the edge of the pool. Write this function in intercept form and find the maximum depth of the pool.

Chef's Tips

- Read each question carefully before answering.
- Use the provided space for calculations and notes.
- Show all work for free response questions.